THE SIMPLICIAL RESOLUTION OF A^{-1}G BY THE SHRIEK FUNCTORS OF THE AUGMENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the opening 'We encourage the reader to read the proof of Lemma simplicial-resolution-Z-site first' (source lines 1945-1947) is a reading suggestion, not an omission - the argument that follows is complete and its conclusion rests on the homotopy equivalence proven in that lemma; the description of the functors g_{n!} used 'without further mention' is established by card L46149, and the augmentation machinery by card L46155 of this batch.

Editorial note. The label lemma-simplicial-resolution-augmentation names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The augmentation datum and its topos morphisms. Same chapter, \label{lemma-augmentation-site} (card L46155 of this batch) and Remark \label{remark-augmentation-site}. Lines 1936-1937: the morphisms a_n, a and the adjunctions a_n^{-1} -| a_{n,*} used to define the maps.
- The component shriek functors and their formulas. Same chapter, \label{lemma-restriction-to-components-site} (card L46149 of this batch). Lines 1937-1943: g_{n!}(a_n^{-1}G) restricted to C_m is the direct sum over varphi : [n] -> [m] of f_varphi^{-1}a_n^{-1}G = a_m^{-1}G.
- The simplicial resolution of the constant sheaf Z. Same chapter, \label{lemma-simplicial-resolution-Z-site}. Lines 1966-1985: the g_m^{-1}-image of the complex is a_m^{-1}G tensor the Z-complex, whose homotopy equivalence with Z in degree 0 was proven there.
- The cosimplicial identities of the coface maps. Simplicial chapter, the cosimplicial identities of the delta^n_i. Lines 1946-1952: the alternating-sum maps d^n_i are adjoints to the coproduct insertions labeled by delta^n_i.

Source reference. Original reference 0D70: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1141–1141. Prerequisite pointer invoked at proof

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line(s) 1948; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 970–970. Prerequisite pointer invoked at proof line(s) 1950; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WI: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1711–1711. Prerequisite pointer invoked at proof line(s) 1947; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D6Z: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1125–1125. Prerequisite pointer invoked at proof line(s) 1932; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 1930; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE let a_0 be an augmentation towards a site $\mathcal{D}$ as in Remark 0D6Z. For any abelian sheaf $\mathcal{G}$ on $\mathcal{D}$ there is an exact complex*

$$\ldots \to g_{2!}(a_2^{-1}\mathcal{G}) \to g_{1!}(a_1^{-1}\mathcal{G}) \to g_{0!}(a_0^{-1}\mathcal{G}) \to a^{-1}\mathcal{G} \to 0$$

of abelian sheaves on $\mathcal{C}_{total}$.

Proof. We encourage the reader to read the proof of Lemma 09WI first. We will use Lemma 0D70 and the description of the functors $g_{n!}$ in Lemma 09WG without further mention. In particular $g_{n!}(a_n^{-1}\mathcal{G})$ is the sheaf on $\mathcal{C}_{total}$ whose restriction to $\mathcal{C}_m$ is the sheaf

$$\bigoplus\nolimits_{\varphi:[n]\to[m]} f_\varphi^{-1} a_n^{-1}\mathcal{G} = \bigoplus\nolimits_{\varphi:[n]\to[m]} a_m^{-1}\mathcal{G}$$

As maps of the complex we take $\sum(-1)^i d_i^n$ where $d_i^n : g_{n!}(a_n^{-1}\mathcal{G}) \to g_{n-1!}(a_{n-1}^{-1}\mathcal{G})$ is the adjoint to the map $a_n^{-1}\mathcal{G} \to \bigoplus_{[n-1]\to[n]} a_n^{-1}\mathcal{G} = g_n^{-1}g_{n-1!}(a_{n-1}^{-1}\mathcal{G})$ corresponding to the factor labeled with $\delta_i^n : [n-1] \to [n]$. The map $g_{0!}(a_0^{-1}\mathcal{G}) \to a^{-1}\mathcal{G}$ is adjoint to the identity map of $a_0^{-1}\mathcal{G}$. Then g_m^{-1} applied to the chain complex in degrees $\ldots, 2, 1, 0$ gives the complex

$$\ldots \to \bigoplus\nolimits_{\alpha\in\mathrm{Mor}_\Delta([2],[m])} a_m^{-1}\mathcal{G} \to \bigoplus\nolimits_{\alpha\in\mathrm{Mor}_\Delta([1],[m])} a_m^{-1}\mathcal{G} \to \bigoplus\nolimits_{\alpha\in\mathrm{Mor}_\Delta([0],[m])} a_m^{-1}\mathcal{G}$$

on $\mathcal{C}_m$. This is equal to $a_m^{-1}\mathcal{G}$ tensored over the constant sheaf $\mathbf{Z}$ with the complex

$$\ldots \to \bigoplus\nolimits_{\alpha\in\mathrm{Mor}_\Delta([2],[m])} \mathbf{Z} \to \bigoplus\nolimits_{\alpha\in\mathrm{Mor}_\Delta([1],[m])} \mathbf{Z} \to \bigoplus\nolimits_{\alpha\in\mathrm{Mor}_\Delta([0],[m])} \mathbf{Z}$$

discussed in the proof of Lemma 09WI. There we have seen that this complex is homotopy equivalent to $\mathbf{Z}$ placed in degree 0 which finishes the proof. □